CALCULUS I - AY 2024-2025

Chapter 1: Functions, Limits, and Continuity

Outline

1 Functions

2 Limits

3 Continuity

Summary

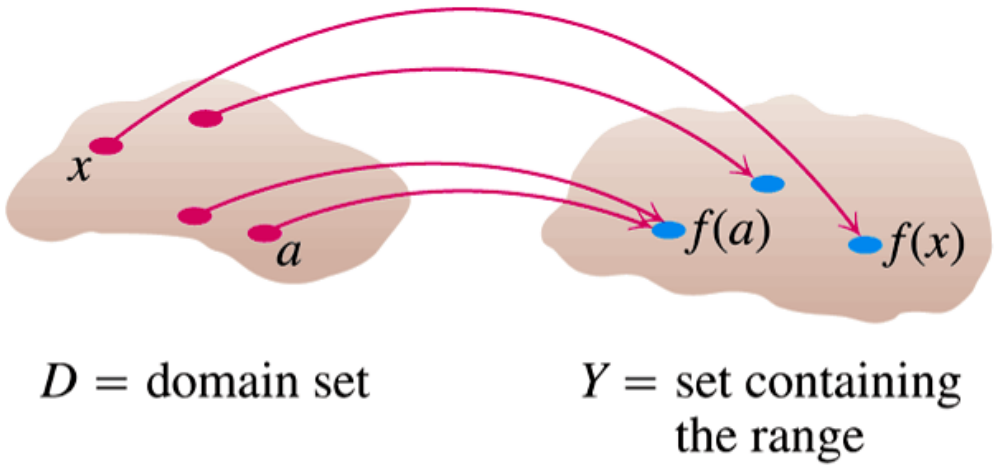
Functions are fundamental to the study of calculus. In this section we study the following topics:

- Functions and their graphs
- Composition of functions
- Important types of functions

• What is a function?

Definition. A function f from a set D to a set Y is a rule that assigns to each element x in D a *unique* (single) element, called $f(x)$, in Y.

The arrow diagram for f looks as follows.



• A function as a kind of machine

The set D of all possible input values is called the domain of the function, while set of all possible values of $f(x)$ as x varies throughout D is called the range of the function.

So we can imagine a function f as a machine which produces an output value $f(x)$ in its range whenever we feed it an input x from its domain.



• How to represent functions?

We will mainly study functions for which the sets D and Y are sets of real numbers.

There are many different ways of describing functions

For example,

- by a description in words (verbally)
- by a table of values (numerically)
- by a graph (visually)
- by an explicit formula (algebraically)

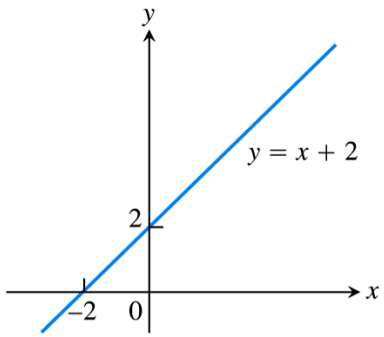
• Visualizing a function

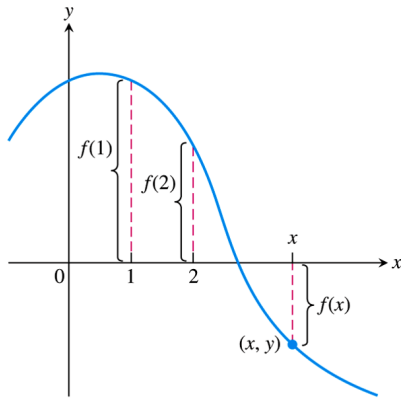
The most common method for visualizing a function is its graph.

The graph of a function

For a function f with domain D, its graph is the set of all points (x, y) in the xy-plane, with $y = f(x)$ and x is in the domain of f. We write

$$G := \{(x, f(x)) : x \in D\}.$$





• Examples

Domain convention

If a function is given by a formula and the domain is not stated explicitly, then the convention is that the domain is the set of all numbers for which the format makes sense and defines a real number.

Exercise

What are the domains of the following functions?

(a) $f(x) = \dfrac{1}{\dfrac{1}{2} - \cos^2 x}$

(b) $g(x) = \dfrac{1}{\sqrt{x-3}} - \sqrt{3-x}$

Answer. (a) $x \neq \dfrac{\pi}{4} + k\dfrac{\pi}{2}$ $(k = 0, \pm 1, \pm 2, ...)$; (b) $x = \emptyset$.

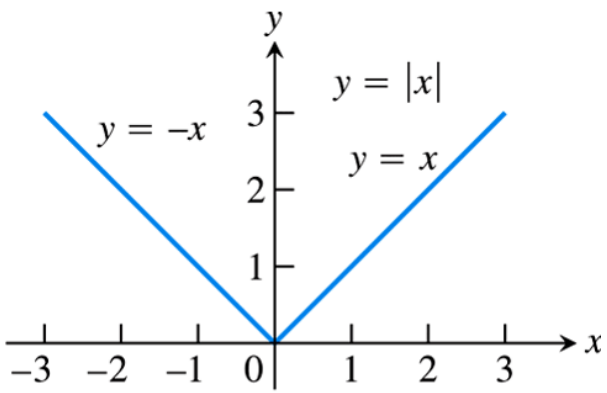
• Piecewise-defined functions

The functions defined by different formulas in different parts of their domains are called piecewise-defined functions.

Example (Absolute value function)

$$|x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0. \end{cases}$$

The graph is as follows.

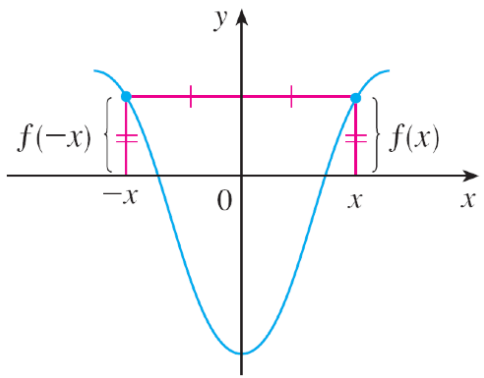


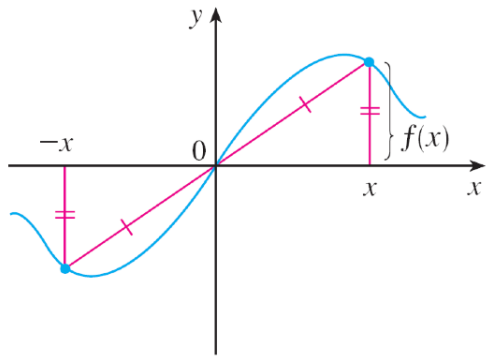
• Even and odd functions: Symmetry

Definition. A function $y = f(x)$ is called

- even, if $f(-x) = f(x)$,
- odd, if $f(-x) = -f(x)$,

for every x in its domain.



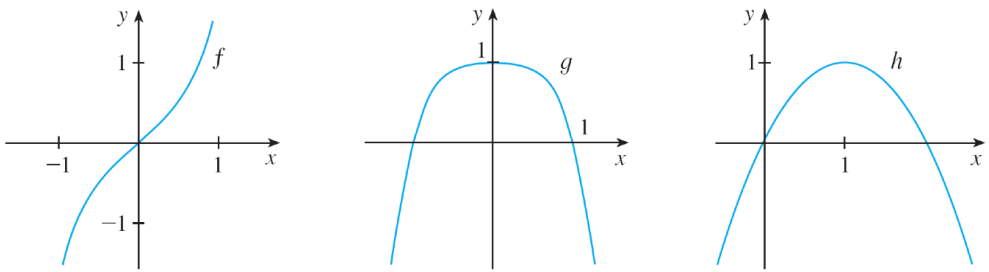


Note. The graph of an even function is symmetric about the y-axis, while the graph of an odd function is symmetric about the origin.

Exercise

Determine whether each of the following functions is even, odd, or neither even nor odd.

$(a)\quad f(x) = x^5 + x \qquad (b)\quad g(x) = 1 - x^4 \qquad (c)\quad h(x) = 2x - x^2.$



Answer. (a) odd; (b) even; (c) neither even nor odd.

• Increasing and decreasing functions

Definition. A function $y = f(x)$ defined on an interval I is called

- increasing, if $f(x_1) < f(x_2)$, for any points $x_1 < x_2$ in I,
- decreasing, if $f(x_1) > f(x_2)$, for any points $x_1 < x_2$ in I.

Note.

- A function is increasing if its graph climbs or rises as you move from left to right.
- A function is decreasing if its graph descents or falls as you move from left to right.

• Composition of functions (combining functions)

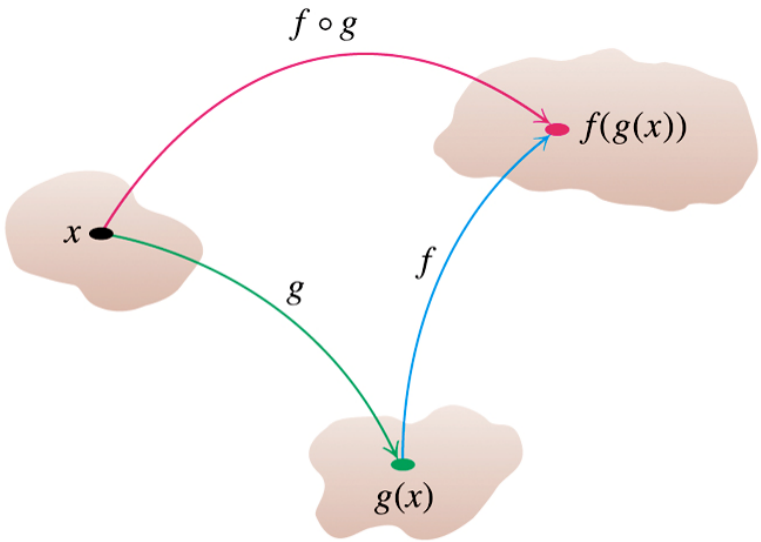
Two functions can be composed at a point x if the value of one function at x lies in the domain of the other.

More precisely, suppose that we have functions $g\colon X \to Y$, $f\colon Y \to Z$, and the range of g is a subset of the domain of f.

Definition. The composite function $f \circ g$ (reading as "f composed with g") is defined by

$$(f \circ g)(x) = f\big(g(x)\big), \text{ for } x \text{ from the domain of } g$$

The arrow diagram for $f \circ g$ looks as follows.



Example

Let $f(x) = x^2 + 1$ and $g(x) = \sqrt{x}$. Determine the function $f \circ g$ and its domain.

Solution. We have

$$(f \circ g)(x) = f(g(x)) = [g(x)]^2 + 1 = (\sqrt{x})^2 + 1 = x + 1.$$

Its domain is $D = \{x \in \mathbb{R} : x \geq 0\}$, which is exactly where the function $g(x)$ is defined.

Question: What is $g \circ f$ and its domain?

Answer. $(g \circ f)(x) = \sqrt{x^2 + 1}$ and $D = (-\infty, \infty)$, as $f(x) \geq 0$, for all x.

Note: The functions $f \circ g$ and $g \circ f$ are not the same.

Compositions can be defined for three or more functions.

For instance, the composite function $f \circ g \circ h$ is found by first applying h, then g, and then f as follows:

$$(f \circ g \circ h)(x) = f\big(g(h(x))\big).$$

◇ **The following section "Important types of functions" is for a self-study (!).**

1. Polynomials
2. Power functions
3. Rational functions
4. Algebraic functions
5. Trigonometric functions
6. Exponential functions
7. Logarithmic functions.

- **Polynomials**

A polynomials is a function of the form

$$P(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_2 x^2 + a_1 x + a_0,$$

where n is a non-negative integer and the numbers a_0, a_1, a_2, $\cdots$, a_n are real constants (called the coefficients of the polynomial).

The domain of any polynomial is $\mathbb{R} = (-\infty, \infty)$. If the leading coefficient $a_n \neq 0$, then n is called the degree of the polynomial.

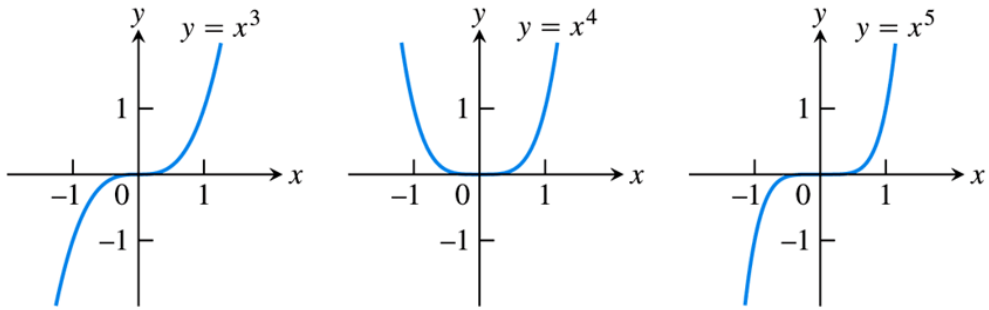
- A polynomial of degree 1 is of the form $P(x) = mx + b$ and it is called a linear function (m is a slope).
- A polynomial of degree 2 is of the form $P(x) = ax^2 + bx + c$ and it is called a quadratic function.
- A polynomial of degree 3 is of the form $P(x) = ax^3 + bx^2 + cx + d$ and it is called a cubic function.

- **Power functions**

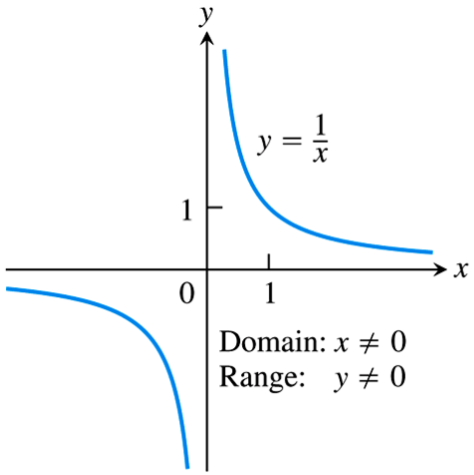
A function $f(x) = x^a$, where a is a constant, is called a power function.

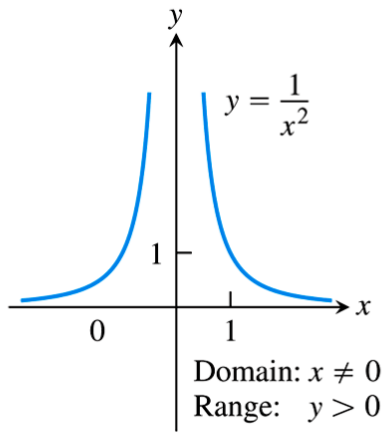
There are several important cases to consider.

- $a = n$, a positive integer

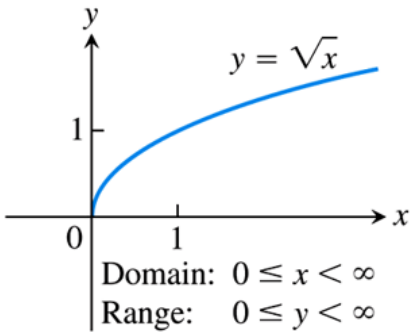


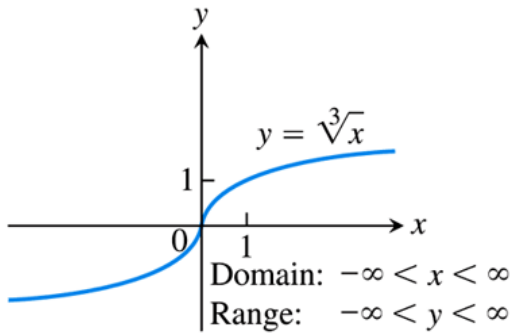
- $a = -1$ or $a = -2$





- $a = \frac{1}{2}, \frac{1}{3}$.





- **Rational functions**

A rational function is a *quotient* or *ratio*

$$f(x) = \frac{P(x)}{Q(x)},$$

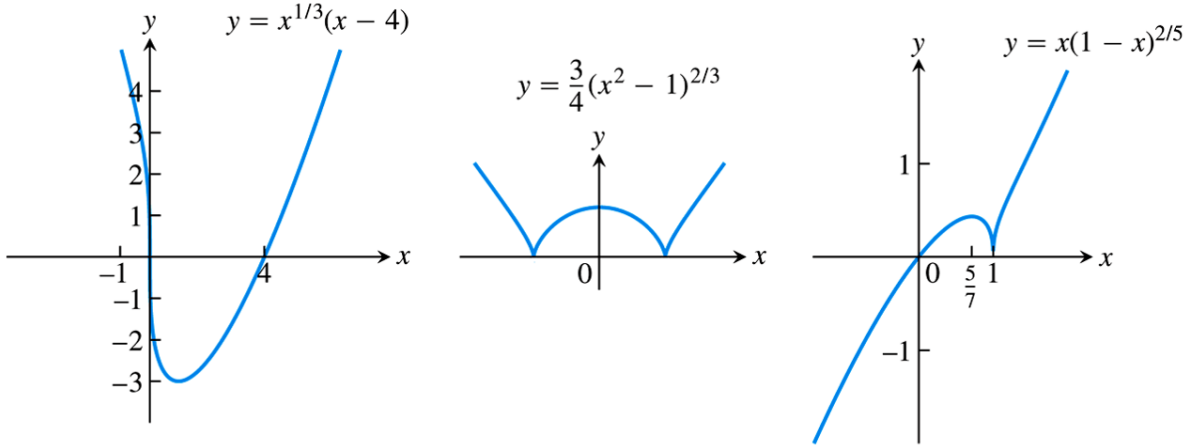
where P and Q are polynomials.

The domain of a rational functions is the set $\{x : Q(x) \neq 0\}$.

- **Algebraic functions**

A function is called an algebraic function if it can be constructed from polynomials using algebraic operations (such as addition, subtraction, multiplication, division, and taking roots).

Figures below display the graphs of some algebraic functions.



- **Trigonometric functions**

Recall that a function $f(x)$ is said to be periodic if there is a positive number T such that

$$f(x+T) = f(x), \text{ for every value of } x.$$

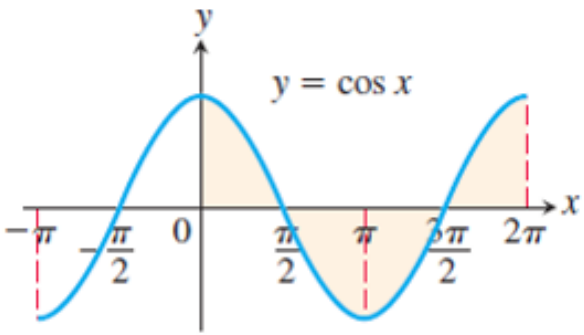
The smallest such value of T is called the period of f.

We have *six* basic trigonometric functions (cosine, sine, tangent, cotangent, secant, and cosecant):

$$\cos x,\ \sin x,\ \tan x,\ \cot x,\ \sec x = \frac{1}{\cos x},\ \csc x = \frac{1}{\sin x}.$$

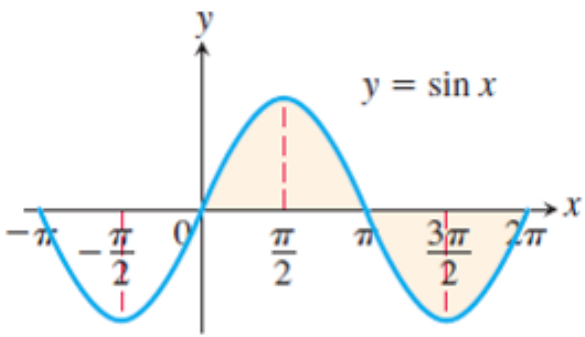
The graphs of these functions are showed below, where the shading for each function indicates its periodicity.

- Functions $\cos x$ and $\sin x$.



Domain: $-\infty < x < \infty$
Range: $-1 \le y \le 1$
Period: 2π

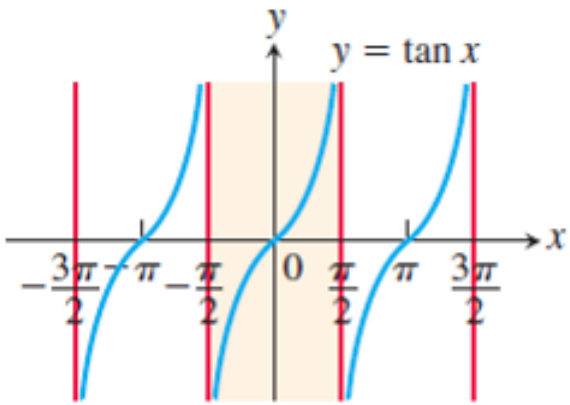
(a)



Domain: $-\infty < x < \infty$
Range: $-1 \le y \le 1$
Period: 2π

(b)

- Functions $\tan x$ and $\sec x$.

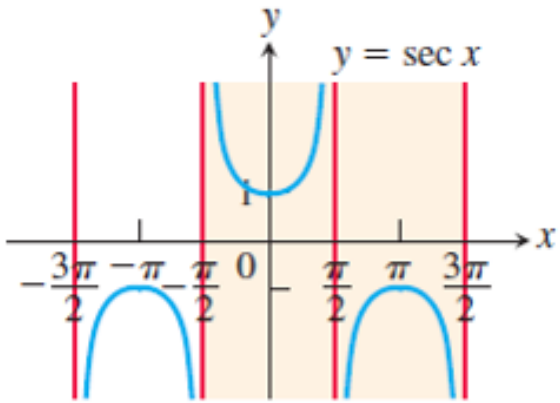


Domain: $x \neq \pm\frac{\pi}{2}, \pm\frac{3\pi}{2}, \ldots$

Range: $-\infty < y < \infty$

Period: π

(c)



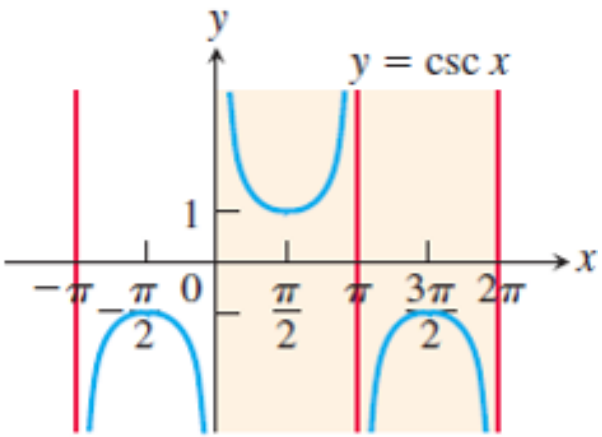
Domain: $x \neq \pm\frac{\pi}{2}, \pm\frac{3\pi}{2}, \ldots$

Range: $y \leq -1$ or $y \geq 1$

Period: 2π

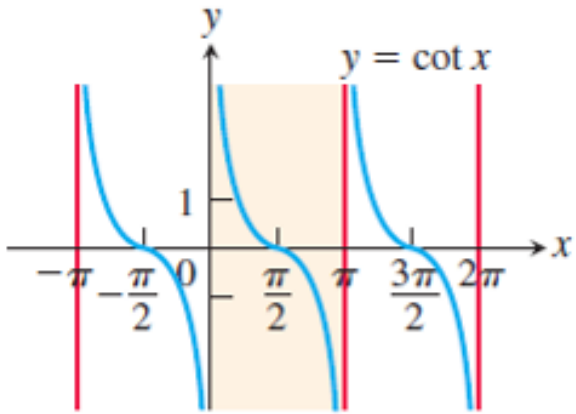
(d)

- Functions csc x and cot x.



Domain: $x \neq 0, \pm\pi, \pm 2\pi, \ldots$
Range: $y \leq -1$ or $y \geq 1$
Period: 2π

(e)



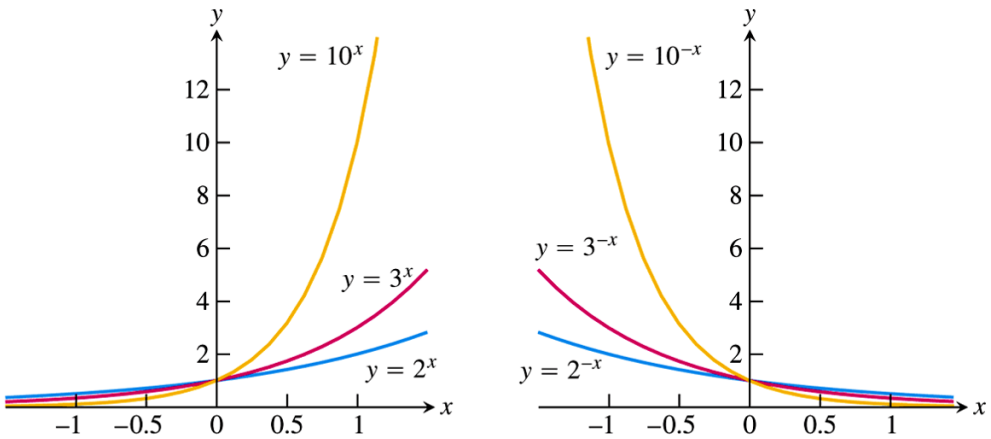
Domain: $x \neq 0, \pm\pi, \pm 2\pi, \ldots$
Range: $-\infty < y < \infty$
Period: π

(f)

- **Exponential functions**

The exponential functions are the functions of the form $f(x) = a^x$, where the base $a > 0$ and $a \neq 1$, is a constant.

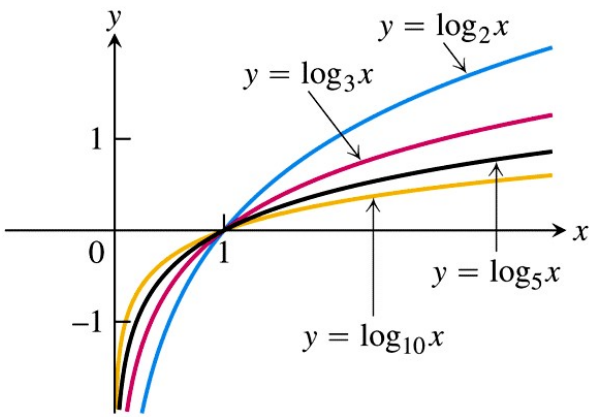
All exponential functions have domain $(-\infty, \infty)$ and range $(0, \infty)$. So these functions never assume the value 0.



- **Logarithmic functions**

The logarithmic functions are the functions of the form $f(x) = log_a x$, where the base $a > 0$ and $a \neq 1$, is a constant. They are the inverse functions of the exponential functions.

The domain is $(0, \infty)$ and the range is $(-\infty, \infty)$.



Outline

1 Functions

2 Limits

3 Continuity

Summary

In this section, we study the following topics:

- Limits, one-sided limits
- Limit laws
- Infinite limits

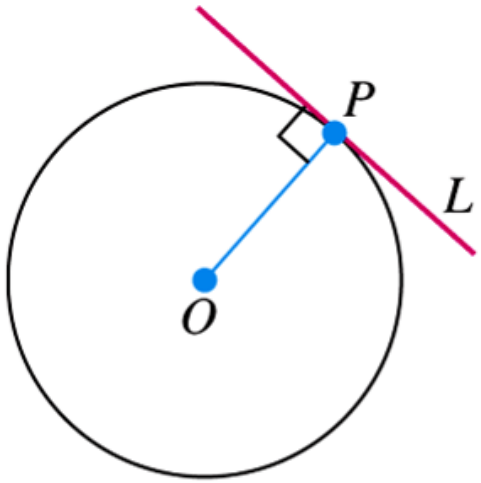
The concept of a limit is a central idea that distinguishes calculus from algebra and trigonometry.

It is fundamental to finding the tangent to a curve.

A tangent to a curve

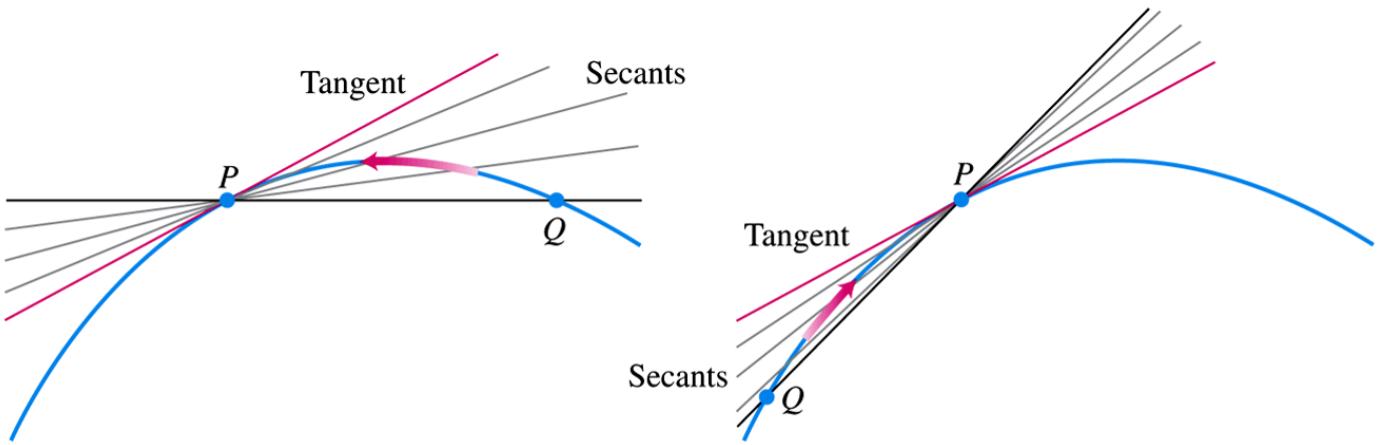
A tangent to a curve is a line that touches the curve.

For a special case, when a curve is a circle, a line is called tangent to a circle if it meets the circle at precisely one point.



• Limit and the tangent to a curve

To define tangency for general curves, we use an approach that analyzes the behavior of the secant lines that pass through P and nearby points Q as Q moves toward P along the curve.



• Limits

In order to solve tangent problems we must be able to find limits.

Limit is one of the most fundamental concepts in calculus (and, in fact, is one of the most important ideas in all of mathematics).

What is a limit?

Let's think about the function

$$f(x) = \frac{\sin x}{x},$$

and focus on how it looks close to the point $x = 0$.

Question. What is the domain of $f(x)$? Answer. $\mathbb{R} \setminus \{0\}$.

Question. What is the value of $f(0)$? Answer. The point 0 does not lie in the domain of f, so f does not assign any value to the point $x = 0$.

The key idea

Even though $f(x) = \dfrac{\sin x}{x}$ has no value defined at the point $x = 0$, we can still look at the values f takes at points close to $x = 0$, and see if those values are approaching some fixed value as x gets closer and closer to 0.

That is precisely what the limit of a function $f(x)$ at a point a is: a number L which the values of the function get "closer and closer" to as the variable x gets "closer and closer" to a.

Let's calculate some values to see what happens as x gets close to 0:



Approaching 0 from the left:

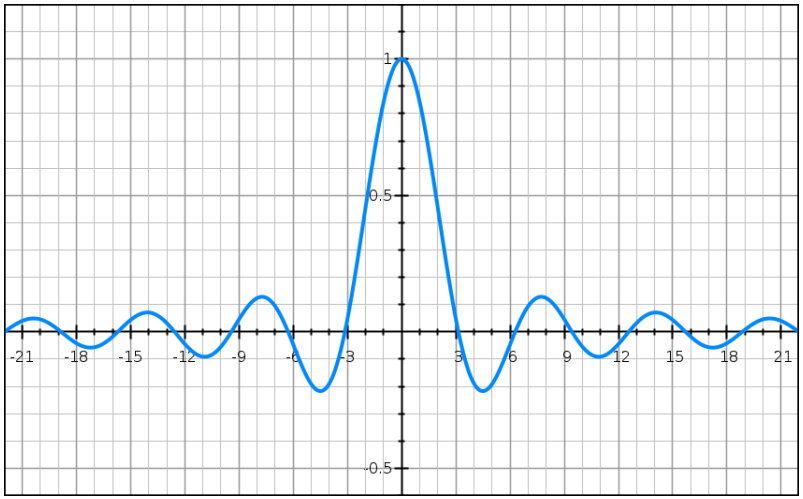
x	-0.25	-0.1	-0.05	-0.01	-0.005	-0.001
$\frac{\sin x}{x}$	0.9896	0.9983	0.9996	0.999983	0.999996	0.999998

Approaching 0 from the right:

x	0.25	0.1	0.05	0.01	0.005	0.001
$\frac{\sin x}{x}$	0.9896	0.9983	0.9996	0.999983	0.999996	0.999998

Comment. We can only guess $\dfrac{\sin x}{x}$ is closer and closer to 1 as x is closer and closer to 0?

The graph of function $f(x) = \dfrac{\sin x}{x}$ is as follows.



• A formal definition of a limit

Definition. Let $f(x)$ be a function defined on some open interval that contains a, except possibly at a itself.

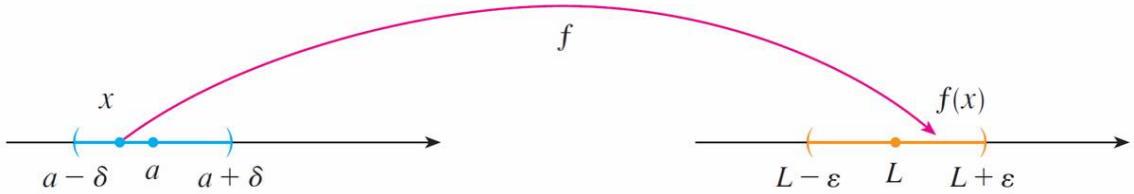
We say that the limit of $f(x)$ as x tends to a is L, and write

$$\lim_{x \to a} f(x) = L,$$

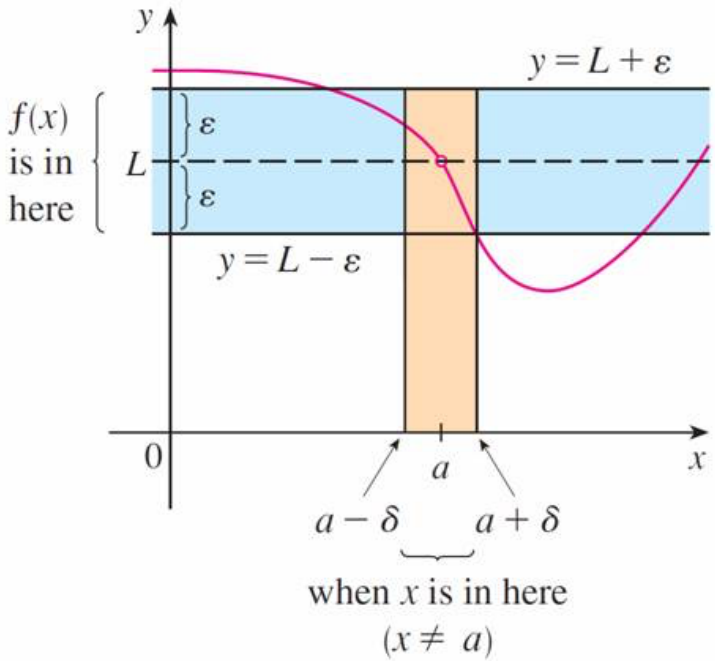
if for every number $\varepsilon > 0$, there exists a corresponding number $\delta > 0$, such that

$$0 < |x - a| < \delta \implies |f(x) - L| < \varepsilon.$$

- We can have the following figure.



- By the graph interpretation, it looks as follows.



• A typical procedure to prove $\lim_{x \to a} f(x) = L$ by definition

To prove a particular limit our task is to explain why we can always choose an appropriate $\delta > 0$, no matter what $\varepsilon > 0$ is.

(1) Let $\varepsilon > 0$ be any number. Choose/find an appropriate number $\delta > 0$.

(This is the important part of the proof)

(Note that δ depends on ε.)

(2) Verify (prove) δ works, that is

$$0 < |x - a| < \delta \implies |f(x) - L| < \varepsilon.$$

Conclude that $\lim_{x \to a} f(x) = L$.

Example

Prove that $\lim_{x \to 1} (7x - 2) = 5$.

Solution.

- Preliminary analysis of the problem (guessing a value for δ).

Let $\varepsilon > 0$ be given. We want to find a number $\delta > 0$ satisfying that

$$0 < |x - 1| < \delta \quad \Longrightarrow \quad \underbrace{|(7x - 2) - 5|}_{7|x-1|} < \varepsilon.$$

So we need to find δ so that

$$\text{if} \quad 0 < |x - 1| < \delta, \quad \text{then} \quad 7|x - 1| < \varepsilon \quad \Longleftrightarrow \quad |x - 1| < \frac{\varepsilon}{7}.$$

To make sure that $|x - 1| < \frac{\varepsilon}{7}$, we may choose for instance $\delta = \frac{\varepsilon}{7}$.

- Proof (showing that this δ works).

Given $\varepsilon > 0$, choose $\delta = \dfrac{\varepsilon}{7}$. In this case, if $0 < |x-1| < \dfrac{\varepsilon}{7}$, then

$$|(7x-2)-5| = 7|x-1| < 7\delta = 7 \cdot \frac{\varepsilon}{7} = \varepsilon.$$

Thus

$$0 < |x-1| < \delta \quad \Longrightarrow \quad |(7x-2)-5| < \varepsilon,$$

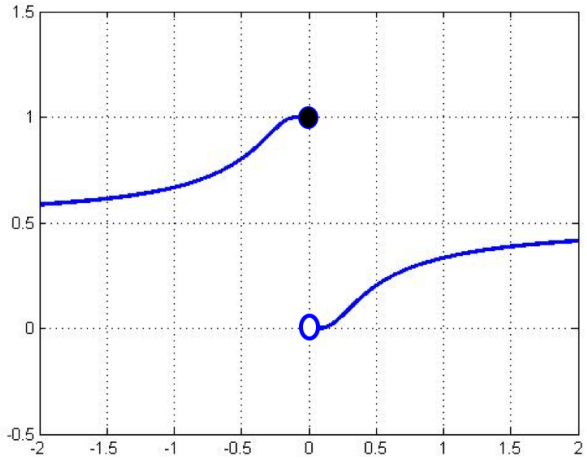
which shows, by the definition of a limit, that $\lim\limits_{x \to 1}(7x-2) = 5$.

Comment. In the solution there were two stages: guessing & proving.

- First we made a preliminary analysis that enabled us to guess a value for δ.
- Next we had to go back and prove in a careful, logical fashion that we had made a correct guess.

• How to show that the limit does not exist?

Let us consider the following function $f(x)$.



How would we use the ε-δ technique to show that the limit $\lim_{x \to 0} f(x)$ does not exist?

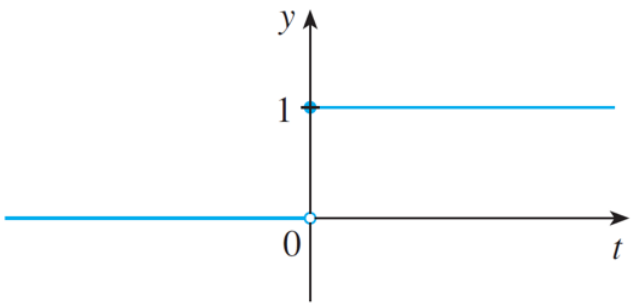
In general, it is not simple. We will come back to this problem later.

One-sided limits

Example

The Heaviside function H is defined by $H(t) = \begin{cases} 0, & t < 0 \\ 1, & t \geq 0. \end{cases}$

This function is named after Oliver Heaviside (1850-1925) (who was an English electrical engineer, mathematician, and physicist) and can be used to describe an electric current that is switched on at time $t = 0$. Its graph is shown below



It is clear that as t tends to 0 from the left, $H(t)$ approaches 0, while as t tends to 0 from the right, $H(t)$ approaches 1.

Moreover, there is no single number that $H(t)$ approaches as t tends to 0. Therefore $\lim_{t\to 0} H(t)$ does not exist.

We indicate the situation symbolically by writing

$$\lim_{\substack{t\to 0\\ t<0}} H(t) = \lim_{t\to 0^-} H(t) = 0,$$

and

$$\lim_{\substack{t\to 0\\ t>0}} H(t) = \lim_{t\to 0^+} H(t) = 1.$$

One-sided limits: ε, δ language

We also have notions of one-sided limits.

Definition of one-side limits

- $\lim_{x \to a^-} f(x) = L$

  if for every number $\varepsilon > 0$ there is a number $\delta > 0$ such that

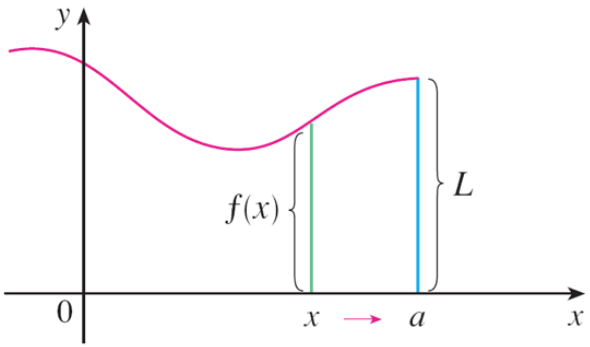
$$a - \delta < x < a \quad \Longrightarrow \quad |f(x) - L| < \varepsilon.$$

- $\lim_{x \to a^+} f(x) = L$

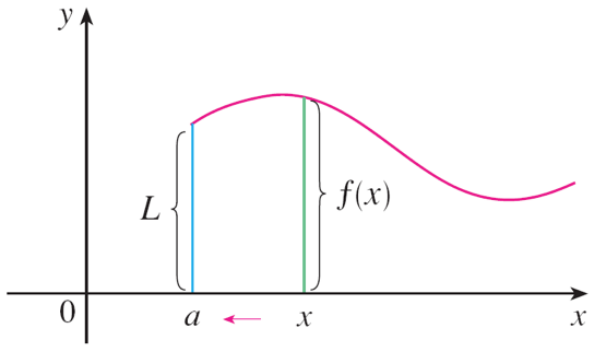
  if for every number $\varepsilon > 0$ there is a number $\delta > 0$ such that

$$a < x < a + \delta \quad \Longrightarrow \quad |f(x) - L| < \varepsilon.$$

The definitions of one-sided limits are illustrated in the following figure.



(a) $\lim_{x \to a^-} f(x) = L$



(b) $\lim_{x \to a^+} f(x) = L$